

SUHYEON YU

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EDUCATION

Rice University

Master of Computer Science (GPA: 3.91/4.00), Korean Graduate Student Association

Houston, TX, USA

Aug. 2025 – Dec. 2026

Sungkyunkwan University

B.S. in Computer Science and Engineering

Suwon, South Korea

Mar. 2018 – Aug. 2024

EXPERIENCE

Software Engineering Intern

Jun. 2026 – Aug. 2026

ABC Fitness

Austin, TX, USA

- **Frontend Modernization:** Rewrote trainer and client profile pages from jQuery to **React** as part of a company-wide modernization initiative for ABC Trainerize, a platform serving **400,000+ fitness professionals** and **1.6M+ clients** worldwide.
- **UI/UX Redesign:** Redesigned the user profile interface (**Claude Design, Figma**) beyond the original migration scope, replacing stacked sections with a card grid and moving profile editing into a modal dialog.
- **Async State Management:** Implemented asynchronous workflows with **RxJS** to coordinate multi-endpoint profile API requests, suppress stale responses, and prevent overlapping operations across profile editing and upload workflows.
- **Unit Testing:** Achieved **97% branch coverage** with **Jest** for profile business logic, including role-based rendering, permission-based editing, and API request and response handling.
- **Incremental Delivery:** Integrated a **LaunchDarkly** feature flag for incremental rollout and split the profile rewrite into smaller, focused pull requests to simplify code review.
- **Cross-Functional Collaboration:** Iterated on profile UI designs with engineers and a UX designer and partnered with a QA analyst to validate and refine behavior.

PROJECT

PyTorch-CKA | Python, PyTorch

- **AI Explainability:** Built an open-source Python library that compares internal representations across layers of **CNNs** and **LLMs**, addressing slow computation and memory buildup in existing CKA tools.
- **Performance Optimization:** Reduced HSIC trace complexity from **$O(B^3)$** to **$O(B^2)$** per layer pair (B: batch size) by replacing full matrix multiplication with Frobenius inner products, achieving an overall **$5.7\times$** speedup over an existing library on H100.
- **Memory Optimization:** Implemented automatic activation cleanup and forward-hook removal after CKA computation, reducing retained GPU memory by **95.7%** compared to an existing library (ResNet-18, batch size 1,000).
- **Deployment & Maintenance:** Published on PyPI (**~400 monthly downloads**) and automated CI testing across Python 3.10–3.14.

Machine Unlearning Comparator | React, TypeScript, D3, Zustand, Python, PyTorch

- Built the frontend end-to-end for an interactive visualization dashboard for evaluating machine unlearning methods across accuracy, efficiency, and privacy using **React, TypeScript, D3**, and **Recharts**; project earned **100+ GitHub stars**.
- Designed the UI in **Figma** for a four-stage evaluation workflow spanning Build, Screen, Contrast, and Attack, around one A/B selection that drives class-wise accuracy, prediction matrices, layer-wise CKA, UMAP embeddings, and MIA simulation.
- Implemented 2,000-point penultimate-layer embedding scatterplots, held interaction latency to **$p95 \leq 11$ ms** via imperative D3 updates and memoization, and cut re-renders **~30%** by moving shared A/B state to **Zustand** selectors.

Online MyeongNyunDang | Next.js, TypeScript, TanStack Query, Redux, Tailwind CSS, Docker, AWS

- Built the frontend for an online education platform for students of my college using **Next.js, TypeScript, TanStack Query**, and **Redux**, which serves **5,000+** students, deployed on **Docker** and **AWS EC2**.
- Designed and implemented the admin and notification pages and migrated the styling system from Styled Components to **Tailwind CSS** to eliminate CSS-in-JS runtime overhead.
- Cut homepage initial load time by **~30%** with next/image lazy loading, caching with **TanStack Query**, and Next.js SSR/SSG.

PUBLICATION

J. Lee, S. Yu, Y. Jang, S. S. Woo, J. Jo. *Unlearning Comparator: A Visual Analytics System for Comparative Evaluation of Machine Unlearning Methods*. *IEEE Transactions on Visualization and Computer Graphics*, vol. 32, no. 3, Mar. 2026.

SKILLS & TOOLS

Languages: TypeScript, JavaScript, Python | **Frontend:** HTML, CSS, React, Next.js, Redux, TanStack Query, Tailwind CSS, SCSS, D3, Zustand
Backend: Node.js, FastAPI, GraphQL, REST APIs | **Databases:** SQL (PostgreSQL, MySQL), NoSQL (MongoDB) | **Testing:** Vitest, Jest, pytest
Cloud & DevOps: AWS (EC2, S3, Lambda), CI/CD (GitHub Actions), Docker, Git, GitHub | **AI tools:** Cursor, Claude Code, Codex, Grok Build